SESSION 9: TIME SERIES ANALYSIS

Modeling and Forecasting Temporal Data in R

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March 2026

Resource: https://cdam.chuka.ac.ke

auto.arima Function

```r
# Automatic ARIMA selection (forecast package)
library(forecast)
# Basic usage on unemployment data
auto_model <- auto.arima(unemploy_ts)
summary(auto_model)
# Control parameters
auto_model2 <- auto.arima(unemploy_ts,
                          max.p = 5, max.q = 5,  # Limit search space
                          max.d = 2, max.D = 1,  # Max differencing
                          seasonal = TRUE,        # Allow seasonal terms
                          stepwise = TRUE,        # Faster stepwise search
                          approximation = FALSE,  # Exact AIC calculation
                          trace = TRUE)           # Show search progress

# Compare with manual selection
AIC(auto_model)
AIC(best_model)  # From previous slide
# auto.arima uses AICc (corrected AIC) by default for small samples
```

auto.arima Function

Recommendation: Use auto.arima for initial model, then refine based on diagnostics

auto.arima Advantages:

- Automatic stationarity testing
- Searches across p,d,q combinations
- Uses AICc for small samples
- Handles seasonal models
- Saves time on manual identification

Forecasting Basics

```r
# Generate forecasts from fitted model
forecast_values <- forecast(arima_model, h = 12)  # 12 months ahead

# Plot forecasts with confidence intervals
plot(forecast_values, main = "Unemployment Forecast",
     xlab = "Year", ylab = "Thousands Unemployed",
     col = "steelblue", shadecols = "lightgray")

# Extract forecast components
forecast_df <- data.frame(
  period = 1:12,
  forecast = as.numeric(forecast_values$mean),
  lo80 = as.numeric(forecast_values$lower[,1]),
  hi80 = as.numeric(forecast_values$upper[,1]),
  lo95 = as.numeric(forecast_values$lower[,2]),
  hi95 = as.numeric(forecast_values$upper[,2])
)
```

Forecasting Basics

```r
# Point forecasts only
point_forecasts <- forecast_values$mean

# Export forecasts
write_csv(forecast_df, "unemployment_forecast.csv")
```

Note: Forecast uncertainty increases with horizon; confidence intervals widen

Forecast Components:

- Point forecast: Expected value at each future time
- Confidence intervals: Range of plausible values
- 80% and 95% intervals typically provided
- Intervals widen as forecast horizon increases

Prediction Intervals

```r
# Forecast with custom confidence levels
forecast_custom <- forecast(arima_model, h = 12, level = c(80, 90, 95))

# Extract intervals
intervals <- forecast_custom$lower  # Lower bounds
intervals_upper <- forecast_custom$upper  # Upper bounds

# Plot with multiple intervals
plot(forecast_custom, main = "Unemployment Forecast with Intervals",
     col = "steelblue", shadecols = c("lightblue", "gray90", "gray95"))

# Interpretation: 95% prediction interval
# "We are 95% confident the true value will fall within this range"

# Note: Prediction intervals assume model is correct and residuals are normal
# In practice, intervals may be too narrow if assumptions violated
```

Prediction Intervals

```r
# Extract specific interval
forecast_custom$lower[,2]  # 95% lower bounds
forecast_custom$upper[,2]  # 95% upper bounds
```

Caution: Prediction intervals do not account for model uncertainty or structural breaks

Understanding Intervals:

- Wider intervals = more uncertainty
- Intervals widen with forecast horizon
- Based on residual variance estimates
- Assume model structure remains valid

Forecast Accuracy Measures

```r
# Split data into training and test sets
train_size <- floor(0.8 * length(unemploy_ts))
train <- unemploy_ts[1:train_size]
test <- unemploy_ts[(train_size + 1):length(unemploy_ts)]

# Fit model on training data
train_model <- auto.arima(train)

# Forecast on test period
test_forecast <- forecast(train_model, h = length(test))

# Calculate accuracy measures
accuracy(test_forecast, test)
# Returns: ME, RMSE, MAE, MPE, MAPE, MASE, ACF1
```

Forecast Accuracy Measures

```
# Key metrics:
# RMSE: Root Mean Squared Error (sensitive to outliers)
# MAE: Mean Absolute Error (more robust)
# MAPE: Mean Absolute Percentage Error (scale-free)
# MASE: Mean Absolute Scaled Error (comparable across series)

# Compare multiple models
accuracy(forecast_ar, test)
accuracy(forecast_ma, test)
```

Best Practice: Use multiple accuracy measures; no single metric tells the whole story

Accuracy Measure Guide:

- RMSE: Penalizes large errors heavily
- MAE: Easier to interpret (same units as data)
- MAPE: Percentage error (good for comparison)
- MASE: Scale-independent, recommended for comparison

Seasonal Decomposition

```r
# Classical decomposition (additive or multiplicative)
decomp_add <- decompose(unemploy_ts, type = "additive")
plot(decomp_add)
# STL decomposition (more robust)
decomp_stl <- stl(unemploy_ts, s.window = "periodic")
plot(decomp_stl)
# Extract components
trend <- decomp_stl$time.series[, "trend"]
seasonal <- decomp_stl$time.series[, "seasonal"]
remainder <- decomp_stl$time.series[, "remainder"]

# Visualize components separately
par(mfrow = c(4, 1))
plot(unemploy_ts, main = "Original: Unemployment")
plot(trend, main = "Trend Component")
plot(seasonal, main = "Seasonal Component")
plot(remainder, main = "Remainder (Irregular)")
par(mfrow = c(1, 1))
```

Seasonal Decomposition

STL: Seasonal-Trend decomposition using Loess; handles any seasonality

STL Advantages:

- Robust to outliers
- Can handle any seasonality period
- Allows seasonal component to change over time
- User-controlled smoothing parameters

For Unemployment:

- Trend shows long-term employment patterns
- Seasonal typically weak in unemployment data
- Remainder captures business cycle effects

Seasonal ARIMA

SARIMA(p,d,q)(P,D,Q)[m]: Extends ARIMA to handle seasonality

```r
# Fit seasonal ARIMA to unemployment
# m = 12 for monthly data with annual seasonality
sarima_model <- auto.arima(unemploy_ts, seasonal = TRUE)
summary(sarima_model)

# Manual SARIMA specification
sarima_manual <- arima(unemploy_ts,
                       order = c(1, 1, 1),        # Non-seasonal part
                       seasonal = list(order = c(0, 1, 1), period = 12))

# Forecast with seasonal model
sarima_forecast <- forecast(sarima_model, h = 24)
plot(sarima_forecast, main = "SARIMA Forecast: Unemployment")

# Interpretation: (0,1,1)[12] means seasonal MA(1) with annual differencing
```

Seasonal ARIMA

SARIMA(p,d,q)(P,D,Q)[m]: Extends ARIMA to handle seasonality

```

# Compare with non-seasonal model
AIC(sarima_model)
AIC(auto_model)
```

Key: Seasonal differencing (D) removes seasonal patterns; seasonal AR/MA terms model seasonal dependence

Seasonal Notation:

- (p,d,q): Non-seasonal ARIMA orders
- (P,D,Q): Seasonal ARIMA orders
- **m** : Seasonal period (12 for monthly, 4 for quarterly)

For Unemployment:

- Seasonality typically weak
- Non-seasonal ARIMA often sufficient
- Test both and compare AIC

Exponential Smoothing

```r
# Simple exponential smoothing (no trend, no seasonality)
ses_model <- ses(unemploy_ts, h = 12)
plot(ses_model, main = "Simple Exponential Smoothing")

# Holt's linear trend method (trend, no seasonality)
holt_model <- holt(unemploy_ts, h = 12)
plot(holt_model, main = "Holt's Linear Trend")

# Holt-Winters method (trend and seasonality)
hw_model <- hw(unemploy_ts, seasonal = "multiplicative", h = 12)
plot(hw_model, main = "Holt-Winters Forecast")

# Automatic selection among exponential smoothing models
ets_model <- ets(unemploy_ts)  # Selects best ETS model
summary(ets_model)
forecast_ets <- forecast(ets_model, h = 12)
plot(forecast_ets)
```

Exponential Smoothing

```
# Compare with ARIMA
accuracy(forecast_ets, test)
accuracy(forecast_arima, test)
```

ETS Framework: Error, Trend, Seasonality components (additive or multiplicative)
Advantage: Exponential smoothing often performs well with less parameter estimation than ARIMA

When to Use:

- Short-term forecasting
- Limited historical data
- When interpretability matters
- As benchmark for complex models

Practical Example: Complete Workflow

```r
# Complete unemployment forecasting workflow

# 1. Load and prepare data
library(forecast)
library(readr)
economics_data <- read_csv("economics.csv")
unemploy_ts <- ts(economics_data$unemploy,
                  start = c(1967, 7), frequency = 12)

# 2. Visualize
plot(unemploy_ts, main = "US Unemployment")

# 3. Check stationarity
adf.test(unemploy_ts)

# 4. Fit model
model <- auto.arima(unemploy_ts)
summary(model)
```

Practical Example: Complete Workflow

```r
# 5. Check diagnostics
tsdiag(model)

# 6. Forecast
forecast_vals <- forecast(model, h = 12)
plot(forecast_vals)

# 7. Export results
write_csv(data.frame(forecast = as.numeric(forecast_vals$mean)),
          "unemployment_forecast.csv")
```

Pro Tip: Document each step; reproducibility is essential for reliable forecasting

Workflow Checklist:

- Save all code in scripts
- Use version control (Git)
- Document data sources and transformations
- Keep raw data separate from analysis

Handling Missing Values

```r
# Introduce missing values for demonstration
ts_missing <- unemploy_ts
ts_missing[sample(1:length(ts_missing), 10)] <- NA

# Method 1: Interpolation (for short gaps)
ts_interp <- na.interp(ts_missing)  # From forecast package

# Method 2: Linear interpolation
ts_linear <- approx(time(ts_missing), as.numeric(ts_missing),
                    xout = time(ts_missing))$y
ts_linear_ts <- ts(ts_linear, start = start(ts_missing),
                   frequency = frequency(ts_missing))

# Method 3: Model-based imputation (advanced)
# Use Kalman smoothing via state space models
# library(imputeTS)
# ts_kalman <- na_kalman(ts_missing, model = "auto")
```

Handling Missing Values

```r
# Check impact on analysis
plot(unemploy_ts, col = "black", main = "Missing Value Handling")
lines(ts_interp, col = "blue", lty = 2)
legend("topright", legend = c("Original", "Interpolated"),
       col = c("black", "blue"), lty = c(1, 2))
```

Caution: Imputation adds uncertainty; report missing data handling in methods

Missing Data Strategies:

- Short gaps: Linear or spline interpolation
- Long gaps: Model-based imputation
- Many missing: Consider alternative models
- Always document approach taken

For economics.csv: No missing values in unemploy variable

Irregular Time Series

Challenge: Observations not equally spaced in time

```r
# Create irregular time series for demonstration
set.seed(123)
irregular_times <- sort(cumsum(rexp(100, rate = 0.1)))
irregular_values <- cumsum(rnorm(100, mean = 0.1, sd = 0.5)) + 10

# Method 1: Aggregate to regular intervals
# Create regular time grid
regular_times <- seq(floor(min(irregular_times)),
                     ceiling(max(irregular_times)), by = 1)

# Aggregate values (e.g., mean) in each interval
regular_values <- sapply(regular_times, function(t) {
  idx <- irregular_times >= t & irregular_times < t + 1
  if(sum(idx) > 0) mean(irregular_values[idx]) else NA
})
```

Irregular Time Series

Challenge: Observations not equally spaced in time

```r
# Convert to ts object (handle NAs)
regular_ts <- ts(regular_values, start = floor(min(irregular_times)),
    frequency = 1)

# Method 2: Use continuous-time models (advanced)
# Consider state space models or Gaussian processes
```

Strategy: Aggregate to regular intervals when possible; use specialized methods for true irregularity

Irregular Data Sources:

- Transaction data (purchases, trades)
- Event logs (system events, user actions)
- Sensor data with transmission gaps
- Medical records (patient visits)

For economics.csv: Regular monthly observations

Multiple Time Series

```r
# Load multiple economic series from economics.csv
pce_ts <- ts(economics_data$pce, start = c(1967, 7), frequency = 12)
pop_ts <- ts(economics_data$pop, start = c(1967, 7), frequency = 12)

# Plot together
plot(unemploy_ts, col = "blue", main = "Economic Indicators",
     ylab = "Value")
lines(pce_ts / 100, col = "red")  # Scale for visualization
legend("topleft", legend = c("Unemployment", "PCE (scaled)"),
       col = c("blue", "red"))

# Cross-correlation to assess lead-lag relationships
ccf(unemploy_ts, pce_ts, main = "Cross-Correlation")
```

Multiple Time Series

```r
# Vector autoregression (VAR) for joint modeling
# install.packages("vars")
library(vars)
var_data <- cbind(unemploy_ts, pce_ts)
var_model <- VAR(var_data, p = 2, type = "const")
summary(var_model)
```

Application: Modeling interdependent series for portfolio analysis, policy evaluation

Multiple Series Applications:

- Portfolio optimization (multiple assets)
- Economic indicators (GDP, inflation, unemployment)
- Regional data (multiple locations)
- Product lines (multiple SKUs)

Cross-Correlation?

Definition: Cross-correlation measures the similarity between two time series as a function of the time lag applied to one of them.

Mathematical Form:

$$\rho_{xy}(k) = \frac{\mathrm{Cov}(X_t, Y_{t+k})}{\sqrt{\mathrm{Var}(X_t)\mathrm{Var}(Y_{t+k})}}$$

where k is the lag (positive or negative).

What It Tells Us:

- **Direction:** Does X lead Y, or does Y lead X?
- **Strength:** How strongly are the series related at each lag?
- **Timing:** At what lag is the relationship strongest?

Key Insight: Cross-correlation helps identify predictive relationships and potential causal pathways between variables.

Applications of Cross-Correlation

Economic Analysis:

- Does unemployment lead inflation (Phillips curve)?
- Do interest rate changes predict GDP growth?
- Leading indicators for business cycles

Environmental Science:

- Temperature anomalies and CO_2 concentrations
- Rainfall patterns and river discharge
- Climate teleconnections (e.g., ENSO effects)

Business and Finance:

- Marketing spend and sales revenue (with lag)
- Stock returns and market indices
- Supply chain lead times and inventory levels

Critical Note: Correlation does not imply causation; use domain knowledge to interpret lead-lag relationships.

Interpreting Cross-Correlation Plots

Reading the CCF Plot:

- **X-axis (Lag):** Time shift applied to the second series
- **Y-axis (CCF):** Correlation coefficient (-1 to +1)
- **Confidence bands:** Dotted lines at $\pm 1.96/\sqrt{n}$

Interpretation Rules:

1. Spike **outside** confidence bands at lag $k > 0$:
   - X leads Y by k periods
   - X may help predict future Y
2. Spike **outside** confidence bands at lag $k < 0$:
   - Y leads X by $|k|$ periods
   - Y may help predict future X
3. No significant spikes: No linear predictive relationship at tested lags

Warning: Raw CCF can be misleading if series are autocorrelated; pre-whitening is essential for valid inference.

Implementation of Cross-Correlation

```r
# Pre-whitening: remove autocorrelation before cross-correlation
# Important: raw CCF can be misleading if series are autocorrelated

# Step 1: Fit ARIMA to unemployment series
x_model <- auto.arima(unemploy_ts)
x_resid <- residuals(x_model)

# Step 2: Filter PCE series through unemployment model
pce_resid <- residuals(Arima(pce_ts, model = x_model))

# Step 3: Compute CCF of residuals (pre-whitened)
ccf(x_resid, pce_resid, main = "Pre-whitened Cross-Correlation",
    ylab = "CCF", xlab = "Lag")

# Interpretation:
# Significant spike at lag k: x leads y by k periods (if k > 0)
# Significant spike at lag -k: y leads x by k periods (if k < 0)
```

Implementation of Cross-Correlation

```r
# Test significance: approximate 95% bounds at +/- 1.96/sqrt(n)
n <- length(x_resid)
abline(h = c(-1.96/sqrt(n), 1.96/sqrt(n)), col = "red", lty = 2)
```

Caution: Correlation does not imply causation; use domain knowledge to interpret lead-lag

Cross-Correlation Interpretation:

- Positive lag: X leads Y (X changes first)
- Negative lag: Y leads X (Y changes first)
- Magnitude: Strength of relationship
- Pre-whitening essential for valid inference

Vector AR Models

VAR(p) Model: Multivariate extension of AR for multiple interdependent series

```r
# Fit VAR model (from vars package)
library(vars)

# Select optimal lag order
var_select <- VARselect(var_data, lag.max = 10, type = "const")
var_select$criteria  # AIC, BIC, HQ, FPE

# Fit VAR with selected order
var_final <- VAR(var_data, p = var_select$selection["AIC(n)"], type = "const")
summary(var_final)

# Impulse response functions: effect of shock to one variable on others
irf_result <- irf(var_final, impulse = "unemploy_ts", response = "pce_ts",
                  n.ahead = 10, ortho = TRUE)
plot(irf_result, main = "Impulse Response: Unemployment shock on PCE")

# Forecast from VAR
```

Common Pitfalls

Avoid These Errors:

- Ignoring non-stationarity: Modeling trending data without differencing
- Over-differencing: Creating artificial patterns by excessive differencing
- Misinterpreting ACF/PACF: Cutoff patterns require careful judgment
- Ignoring seasonality: Fitting non-seasonal models to seasonal data
- Overfitting: Selecting overly complex models that fit noise
- Ignoring residual diagnostics: Assuming model is adequate without checking
- Extrapolating too far: Forecasts become unreliable at long horizons
- Forgetting structural breaks: Models assume stable relationships over time

Golden Rule: Simpler models often forecast better than complex ones; validate with out-of-sample testing

Additional Warnings:

- Don't ignore domain knowledge
- Beware of spurious correlations
- Account for calendar effects (trading days, holidays)

Best Practices

Before Modeling:

1. Visualize data to identify trend, seasonality, outliers
2. Test for stationarity; difference or transform as needed
3. Consider domain knowledge about data generating process

During Modeling:

1. Start simple (ARIMA(0,1,0) or exponential smoothing)
2. Use auto.arima for initial model, then refine manually
3. Check residuals thoroughly; iterate if patterns remain
4. Compare multiple models using AIC/BIC and out-of-sample accuracy

After Modeling:

1. Report model orders, coefficients, and diagnostics
2. Present forecasts with prediction intervals
3. Acknowledge limitations and uncertainty
4. Update models periodically as new data arrives

Resources for Learning

Books:

- Hyndman and Athanasopoulos (2021). Forecasting: Principles and Practice (free online)
- Box, Jenkins, Reinsel (2015). Time Series Analysis: Forecasting and Control
- Shumway and Stoffer (2017). Time Series Analysis and Its Applications

R Packages:

- forecast: ARIMA, exponential smoothing, automatic modeling
- tseries: Stationarity tests, basic time series functions
- vars: Vector autoregression for multivariate series
- rugarch: GARCH models for financial volatility
- fable: Modern tidyverse-compatible forecasting framework

Online:

- OTexts: https://otexts.com/fpp3/ (free textbook)
- R Time Series Task View: https://cran.r-project.org/web/views/TimeSeries.html
- Forecasting Principles: https://forecastingprinciples.com

Practice Exercises

Try these with economics.csv:

1. Load unemploy variable and convert to ts object
2. Decompose unemployment series using STL; interpret components
3. Fit ARIMA models; compare AIC for different orders
4. Forecast next 12 months; evaluate accuracy
5. Apply exponential smoothing; compare with ARIMA
6. Analyze cross-correlation between unemploy and pce

Challenge:

- Implement rolling window forecasting to assess stability
- Compare forecast accuracy of ARIMA vs ETS vs machine learning
- Handle a time series with missing values using multiple imputation
- Model multiple economic indicators with VAR

Tip: Start with simple examples; gradually increase complexity as understanding grows

Built-in Datasets for Additional Practice:

Summary: Key Concepts

Time Series Fundamentals:

- Temporal dependence requires specialized methods
- Decompose into trend, seasonality, cycle, irregular components
- Stationarity is foundational; transform non-stationary data

Modeling Framework:

- ARIMA(p,d,q) combines autoregression, differencing, moving average
- ACF/PACF guide model identification; AIC/BIC guide selection
- Diagnostics ensure residuals are white noise

Forecasting:

- Point forecasts plus prediction intervals quantify uncertainty
- Accuracy measures (RMSE, MAE, MAPE) evaluate performance
- Simpler models often outperform complex ones in practice

Remember: Time series analysis is iterative; combine statistical rigor with domain expertise

For Unemployment Data:

Summary: R Workflow

```r
# 1. Load and visualize data
library(forecast)
library(readr)
economics_data <- read_csv("economics.csv")
unemploy_ts <- ts(economics_data$unemploy,
                  start = c(1967, 7), frequency = 12)
plot(unemploy_ts)

# 2. Check stationarity and transform if needed
adf.test(unemploy_ts)
if(non_stationary) unemploy_ts <- diff(unemploy_ts)

# 3. Identify model orders using ACF/PACF
acf(unemploy_ts); pacf(unemploy_ts)

# 4. Fit candidate models and compare
model1 <- auto.arima(unemploy_ts)
AIC(model1)

```

Advanced Topics Preview

Beyond Basic ARIMA:

- State space models and Kalman filtering
- Dynamic regression with external predictors
- Structural time series models
- Machine learning hybrids (ARIMA + neural networks)
- Bayesian time series modeling

Specialized Applications:

- High-frequency financial data (intraday, tick data)
- Spatial-temporal modeling (geographic time series)
- Functional time series (curves observed over time)
- Causal inference with time series (intervention analysis)

Next Step: Master fundamentals before advancing; each topic builds on core concepts

Emerging Trends:

- Deep learning for time series (LSTM, Transformer models)

Troubleshooting Checklist

If Model Fails or Forecasts Poorly:

1. Verify data is properly formatted as ts object with correct frequency
2. Confirm stationarity; check if differencing is adequate
3. Examine residuals for remaining patterns or heteroscedasticity
4. Try alternative model specifications (different p,d,q orders)
5. Consider transformations (log, Box-Cox) for variance stabilization
6. Check for outliers or structural breaks affecting estimation
7. Validate with out-of-sample testing, not just in-sample fit
8. Consult domain experts about plausible data generating mechanisms

Pro Tip: When stuck, simplify: start with naive forecast, then add complexity incrementally

Common Error Messages:

- "non-stationary AR part": Increase differencing (d)
- "non-invertible MA part": Reduce MA order (q)
- "convergence failed": Try different starting values or simpler model

Questions?

Common Topics:

- Choosing between ARIMA and exponential smoothing
- Interpreting AIC differences between models
- Handling multiple seasonalities (e.g., daily + weekly)
- Forecasting with limited historical data
- Incorporating external variables into forecasts

No question is too basic - time series analysis has many nuances

Dataset-Specific Questions:

- How to handle economics.csv data structure
- Best model for unemployment forecasting
- How far ahead can we forecast reliably

Next Steps

After Mastering This Session:

1. Practice with diverse datasets (economic, environmental, business)
2. Explore the fable package for tidy forecasting workflows
3. Learn Bayesian time series with rstan or brms
4. Study causal inference methods for time series (DID, synthetic control)
5. Contribute to open-source time series packages

Real-World Application:

- Start a forecasting project with your own data
- Document assumptions, methods, and limitations clearly
- Update models regularly as new data arrives
- Communicate uncertainty effectively to stakeholders

Remember: Good forecasting combines statistical skill, domain knowledge, and clear communication

Learning Path:

References

Core Textbooks:

- Hyndman, R.J., & Athanasopoulos, G. (2021). Forecasting: Principles and Practice (3rd ed.)
- Box, G.E.P., Jenkins, G.M., Reinsel, G.C. (2015). Time Series Analysis: Forecasting and Control
- Shumway, R.H., & Stoffer, D.S. (2017). Time Series Analysis and Its Applications

Key Papers:

- Makridakis, S., et al. (2018). The M4 Competition
- Hyndman, R.J., & Khandakar, Y. (2008). Automatic Time Series Forecasting
- Petropoulos, F., et al. (2022). Forecasting: Theory and Practice

R Documentation:

- forecast package: https://pkg.robjhyndman.com/forecast/
- fable package: https://fable.tidyverse.org/
- R Time Series Task View: https://cran.r-project.org/web/views/TimeSeries.html

Thank You!

Questions?

Contact: cdam@chuka.ac.ke
Resources: https://cdam.chuka.ac.ke
Code: https://github.com/cdam-chuka/r-sessions

Next Session: Advanced Topics in Statistical Modeling
Preparation: Review ARIMA concepts; practice with economics.csv
Dataset: economics.csv (unemploy variable for time series)